

What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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TIME-DOMAIN ASTROPHYSICS

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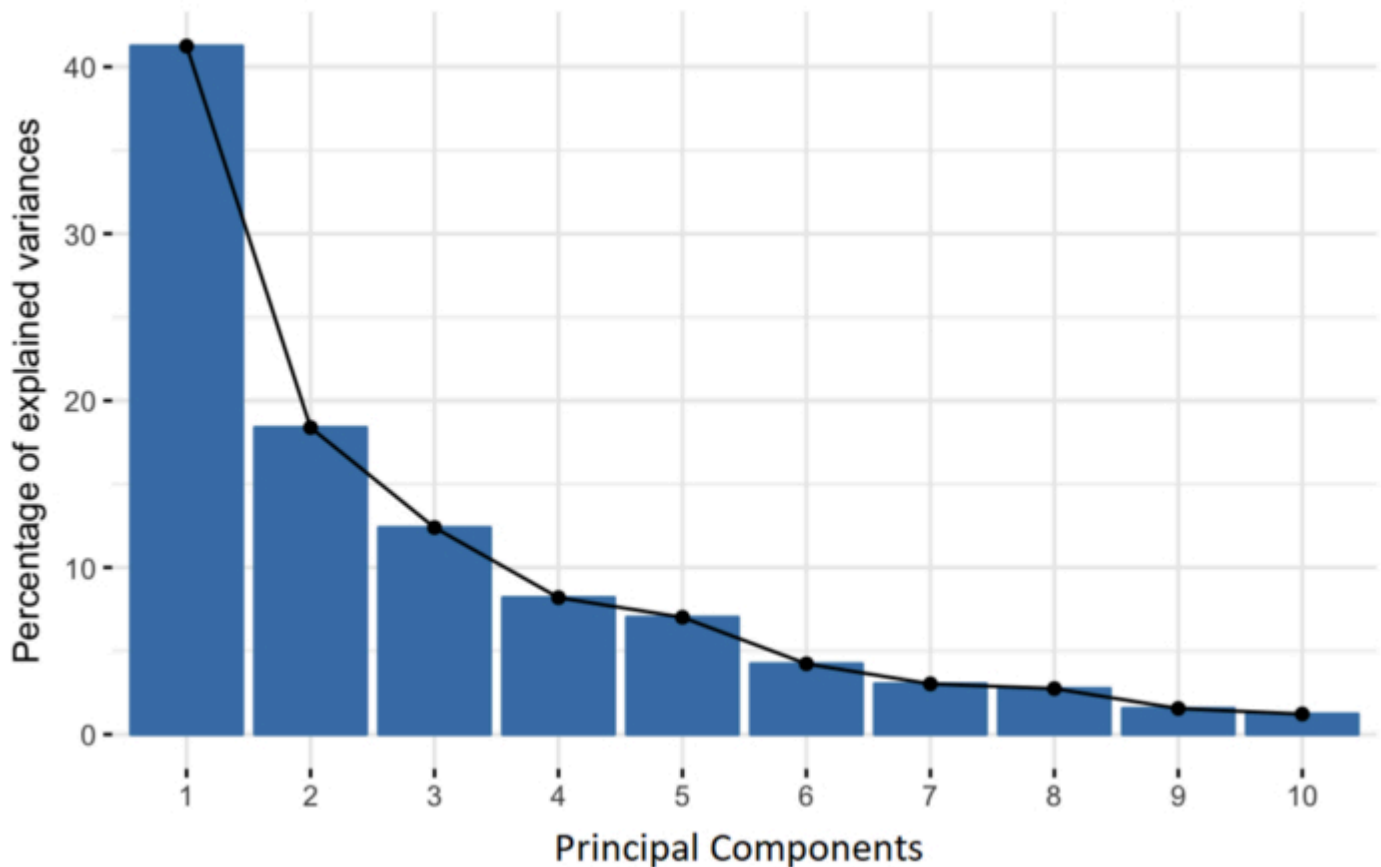


Principal Component Analysis (PCA)

- Principal component analysis (PCA) is a dimensionality reduction technique that transforms a data set into a set of orthogonal components, called *principal components*, which capture the maximum variance in the data.
- PCA simplifies complex data sets while preserving their most important structures.

What Are Principal Components?

- Principal components are new variables that are constructed as linear combinations or mixtures of the initial variables.
- These combinations are done in such a way that the new variables (i.e., principal components) are uncorrelated and most of the information within the initial variables is squeezed or compressed into the first components.
- So, the idea is 10-dimensional data gives you 10 principal components, but PCA tries to put maximum possible information in the first component, then maximum remaining information in the second and so on, until having something like shown below:



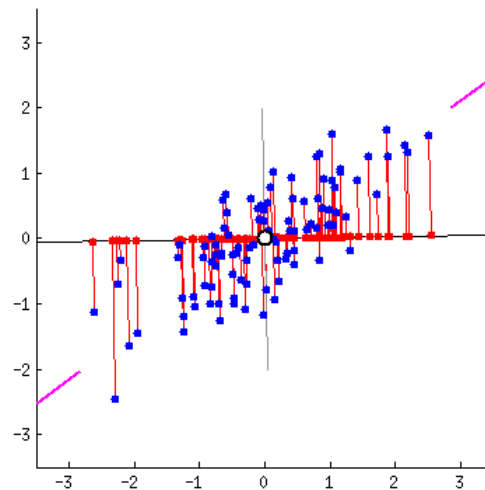
- Organizing information in principal components will allow one to reduce dimensionality without losing much information, and this by discarding the components with low information and considering the remaining components as your new variables.
- Principal components are often less interpretable and do not (necessarily) have any real physical meaning since they are constructed as linear combinations of the initial variables.
- Geometrically speaking, principal components represent the directions of the data that explain a maximal amount of variance.

To put all this simply, just think of principal components as new axes that provide the best angle to see and evaluate the data, so that the differences between the observations are better visible.

- For example, let us assume that the scatter plot of our data set is as shown below, the first principal component is approximately the line that matches the purple marks because it goes

through the origin and it's the line in which the projection of the points (red dots) is the most spread out.

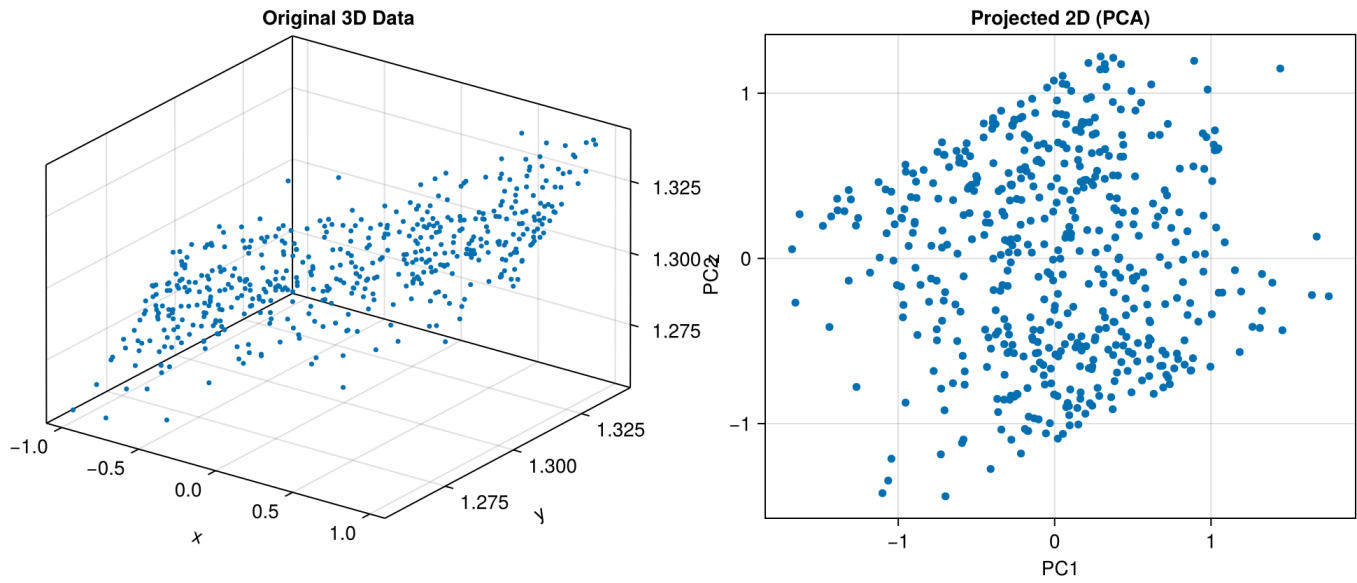
- Or mathematically speaking, it's the line that maximizes the variance (the average of the squared distances from the projected points (red dots) to the origin).



- The second principal component is calculated in the same way, with the condition that it is uncorrelated with (i.e., perpendicular to) the first principal component and that it accounts for the next highest variance.
- This continues until a total of p principal components have been calculated, equal to the original number of variables.

Exercise: dimensionality reduction and pattern exploration

- Let's create a random 3D dataset and look for the 2D description defined by its tow first PCs.



How Principal Component Analysis Works: 5 Steps

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- Principal component analysis can be broken down into five steps. We'll go through each step, providing logical explanations of what PCA is doing.

Step 1: Standardization and Centering Data

- The aim of this step is to standardize the range of the continuous initial variables so that each one of them contributes equally to the analysis.
- More specifically, the reason why it is critical to perform standardization prior to PCA, is that the latter is quite sensitive regarding the variances of the initial variables. That is, if there are large differences between the ranges of initial variables, those variables with larger ranges will dominate over those with small ranges (for example, a variable that ranges between 0 and 100 will dominate over a variable that ranges between 0 and 1), which will lead to biased results. So, transforming the data to comparable scales can prevent this problem.
- Mathematically, this can be done by subtracting the mean and dividing by the standard deviation for each value of each variable. E.g., if x is the considered variable, the standardized variable, z , turns out to be:

$$z = \frac{x - \langle x \rangle}{\sigma_x}$$

Step 2: Covariance Matrix Computation

- The aim of this step is to understand how the variables of the input data set are varying from the mean with respect to each other, or in other words, to see if there is any relationship between them. Because sometimes, variables are highly correlated in such a way that they contain redundant information. So, in order to identify these correlations, we compute the covariance matrix.
- The covariance matrix is a $p \times p$ symmetric matrix (where p is the number of dimensions) that has as entries the covariances associated with all possible pairs of the initial variables. For example, for a 3-dimensional data set with 3 variables x , y , and z , the covariance matrix is a 3×3 data matrix of this form:

$$\begin{pmatrix} Cov(x, x) & Cov(x, y) & Cov(x, z) \\ Cov(y, x) & Cov(y, y) & Cov(y, z) \\ Cov(z, x) & Cov(z, y) & Cov(z, z) \end{pmatrix}$$

- Since the covariance of a variable with itself is its variance, $Cov(a, a) = Var(a)$, in the main diagonal we actually have the variances of each initial variable. And since the covariance is commutative, $Cov(a, b) = Cov(b, a)$, the entries of the covariance matrix are symmetric with respect to the main diagonal, which means that the upper and the lower triangular portions are equal.

Step 3: Eigen Decomposition and Identifying Principal Components

- Eigenvectors and eigenvalues are the linear algebra concepts that we need to compute from the covariance matrix in order to determine the principal components of the data.
- The principal components are often computed by eigendecomposition of the data covariance matrix or singular value decomposition of the data matrix.
- The eigenvectors of the Covariance matrix are actually the directions of the axes where there is the most variance (most information) and that we call Principal Components. And eigenvalues are simply the coefficients attached to eigenvectors, which give the amount of variance carried in each Principal Component.
- By ranking your eigenvectors in order of their eigenvalues, highest to lowest, you get the principal components in order of significance.
- After having the principal components, to compute the percentage of variance (information) accounted for by each component, we divide the eigenvalue of each component by the sum of eigenvalues.

Step 3a: Eigendecomposition

- In linear algebra, eigendecomposition (a.k.a. eigenvalue decomposition or EVD) is the factorization of a matrix into a canonical form, whereby the matrix is represented in terms of its eigenvalues and eigenvectors.
- Only diagonalizable matrices can be factorized in this way. When the matrix being factorized is a normal or real symmetric matrix, the decomposition is called "spectral decomposition", derived from the spectral theorem.

Step3b: Singular value decomposition

- In linear algebra, the singular value decomposition (SVD) is a factorization of a real or complex matrix into a rotation, followed by a rescaling followed by another rotation.
- It generalizes the eigendecomposition of a square normal matrix with an orthonormal eigenbasis to any $m \times m$ matrix.

Step 4: Feature Selection and Creating a Feature Vector

- As we saw in the previous step, computing the eigenvectors and ordering them by their eigenvalues in descending order, allow us to find the principal components in order of significance.
- We can also choose whether to keep all these components or discard those of lesser significance (of low eigenvalues), and form with the remaining ones a matrix of vectors that we call Feature vector, e.g. $\mathbf{F}$. This is also seen as *dimensionally reduction*.
- It's up to you to choose whether to keep all the components or discard the ones of lesser significance.
 - A simpler description of your data with fewer variables can justify the loss of information (unless a PC has an exactly null eigenvalue) or not, depending on the specific problem you are dealing with.

Step 5: Data Projection

- In the previous steps, apart from standardization, you do not make any changes on the data, you just select the principal components and form the feature vector, but the input data set remains always in terms of the original axes (i.e., in terms of the initial variables).
- In this step, which is the last one, the aim is to use the feature vector formed using the eigenvectors of the covariance matrix, to reorient the data from the original axes to the ones represented by the principal components (hence the name Principal Components Analysis). This can be done by multiplying the transpose of the original data set by the transpose of the feature vector:

$$\mathbf{x}_{new} = \mathbf{F}^T * \mathbf{z}^T$$

Principal Component Analysis Variants and Extensions

- While classical principal component analysis assumes linearity and orthogonal components, real-world data doesn't always follow these assumptions. This has led to various types of PCA extensions being developed to address specific data challenges.
- Here are some popular PCA variants to know:

1. Kernel PCA (KPCA)

- Kernel PCA extends classical PCA to capture non-linear structures in data. It maps the input data into a high-dimensional feature space using a kernel function, and then performs PCA in this new space. KPCA allows users to uncover complex manifolds in data that linear PCA cannot detect, making it useful in image processing, pattern recognition and non-linear clustering.
 - Key advantage: Captures non-linear relationships between features.

2. Sparse PCA (SPCA)

- Sparse PCA applies sparse constraints (limits to the number active parameters) to principal components, resulting in eigenvectors or other loadings to contain many zeroes. This makes principal components more interpretable and robust in high-dimensional scenarios where the number of features is greater than the number of samples, such as in text mining or genomics.
 - Key advantage: Creates interpretable components and mitigates overfitting in high-dimensional data.

3. Robust PCA (RPCA)

- Robust PCA addresses the sensitivity of classical PCA to outliers by decomposing the data matrix into a low-rank matrix and a sparse matrix that captures the anomalies. This makes RPCA effective for tasks such as fraud detection or background subtraction in video sequences.
 - Key advantage: Handles outliers and noise by separating them from the true low-dimensional structure.

4. Incremental PCA (IPCA)

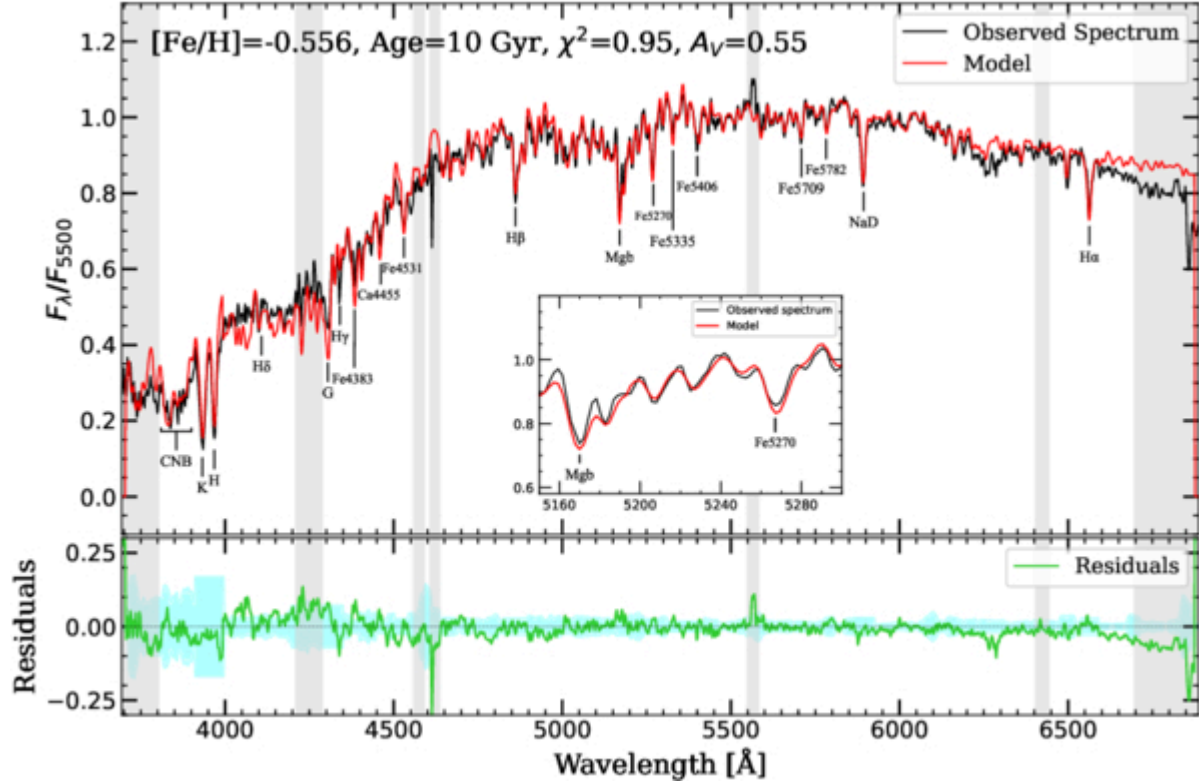
- Incremental PCA is designed for scenarios where a data set is too large to fit into computer memory. As such, IPCA updates the principal components incrementally as new data arrives, making it suitable for data streaming or real-time applications.
 - Key advantage: Scalable to large data sets and suitable for real-time data tasks.

5. Probabilistic PCA (PPCA)

- Probabilistic PCA reformulates PCA as a probabilistic latent variable model, which assumes Gaussian noise and enables full Bayesian treatment. PPCA facilitates principled handling of missing data and uncertainty estimation of data, making it useful in machine learning and statistical modeling.
 - Key advantage: Provides a probabilistic interpretation of data and supports inference under uncertainty.

Exercise: Spectral indices for Galactic Globular Clusters

- This is an exercise based on an actual work published in [Covino et al. \(1996\)](#).
- A set of measurements of “spectral indices” for Globular Clusters was obtained after an observational campaign with the [ESO 1.5m spectroscopic telescope](#).
- “Spectral indices” (pay attention that this term is used in astronomy with different meaning in different contexts) are measurements of the intensities of absorption lines in low-resolution spectra.



- The image above is from [López-Santamaría et al. 2025](#)
- Globular Clusters are star clusters with millions of gravitationally bound members.
 - These stars share (almost) the same chemical composition and age and are all at virtually the same distance from us: they are formidable natural laboratories of stellar evolution.
- Galactic globular clusters show a very old age, comparable to the age of the Universe, but younger globular clusters are observed in other galaxies.



- The globular cluster in the image above is M2.
- Many possible spectral indices can be defined and, roughly speaking, they are mainly sensitive to the global metal content (i.e. the *metallicity*) and age.
- We want now to check whether we can synthesise the information content of our dataset reducong its dimensionality for visualization and easier classification.
- Let's look at the input data, a matrix with several measured spectral indices for a few Galactic GCs:

mydata =

	Obj	CH	MGb	Fe52	Fe53	NaD	Hbeta	CN	altro
1	"NGC104"	0.15	0.106	0.063	0.054	0.089	0.066	0.013	
2	"NGC362"	0.109	0.002	0.038	0.027	0.039	0.066	-0.042	
3	"NGC5024"	missing	0.027	missing	0.013	missing	0.111	missing	
4	"NGC5824"	0.049	0.003	0.039	0.029	0.073	0.101	-0.091	
5	"NGC5946"	0.096	0.019	0.044	0.04	0.129	0.082	-0.097	
6	"NGC6171"	missing	0.058	0.082	0.038	missing	0.057	missing	
7	"NGC6218"	missing	0.06	0.032	0.024	missing	0.049	missing	
8	"NGC6284"	0.059	0.038	0.043	0.031	0.088	0.115	-0.082	
9	"NGC6293"	0.044	-0.004	0.028	0.016	0.097	0.105	-0.106	
10	"NGC6356"	missing	0.101	0.068	0.032	missing	0.067	missing	
	altro								
18	"NGC7099"	0.027	0.013	0.025	0.022	0.043	0.12	-0.115	

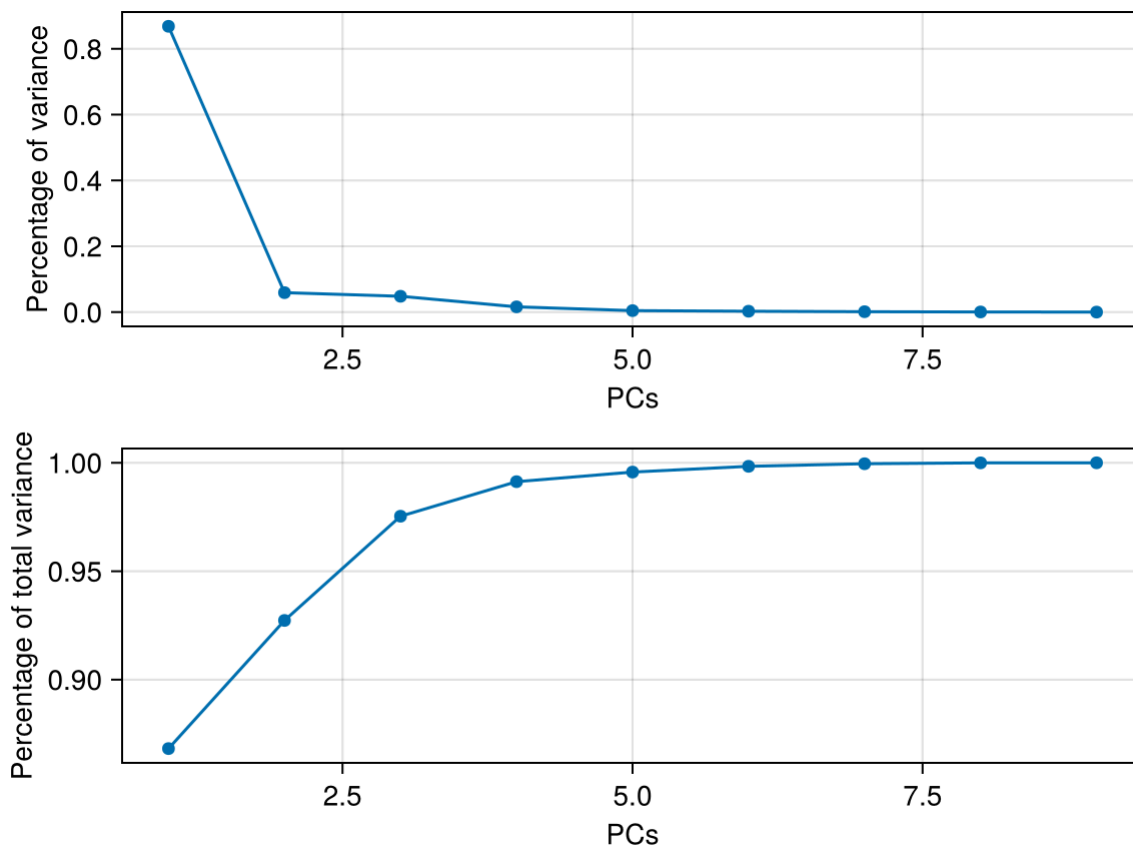
```
1 mydata = read_wsv_with_missing("GC_dataset.csv")
```

- Some entries are missing and therefore we now filter the matrix keeping only the fully complete entries.

```
1 begin
2     dfc = dropmissing(mydata)
3     Xs = Matrix(dfc[:,2:10])
4 end;
```

- First of all, we analyse the eigenvalues of the correlation matrix and the associated variance.

```
1 begin
2     Ms = fit(PCA, Xs'; pratio=1)
3     tvar = principalvars(Ms)/var(Ms)
4     tvartot = [sum(tvar[1:i]) for i in 1:length(tvar)]
5 end;
```



- It is clear that with just two (three) PCs we can describe more than 95% (98%) of the total variance of the sample.
- Let's plot our dataset to the PC space with 2 or 3 dimensions.

```
Mss = PCA(indim = 9, outdim = 3, principalratio = 0.9753738714337457)
```

Pattern matrix (unstandardized loadings):

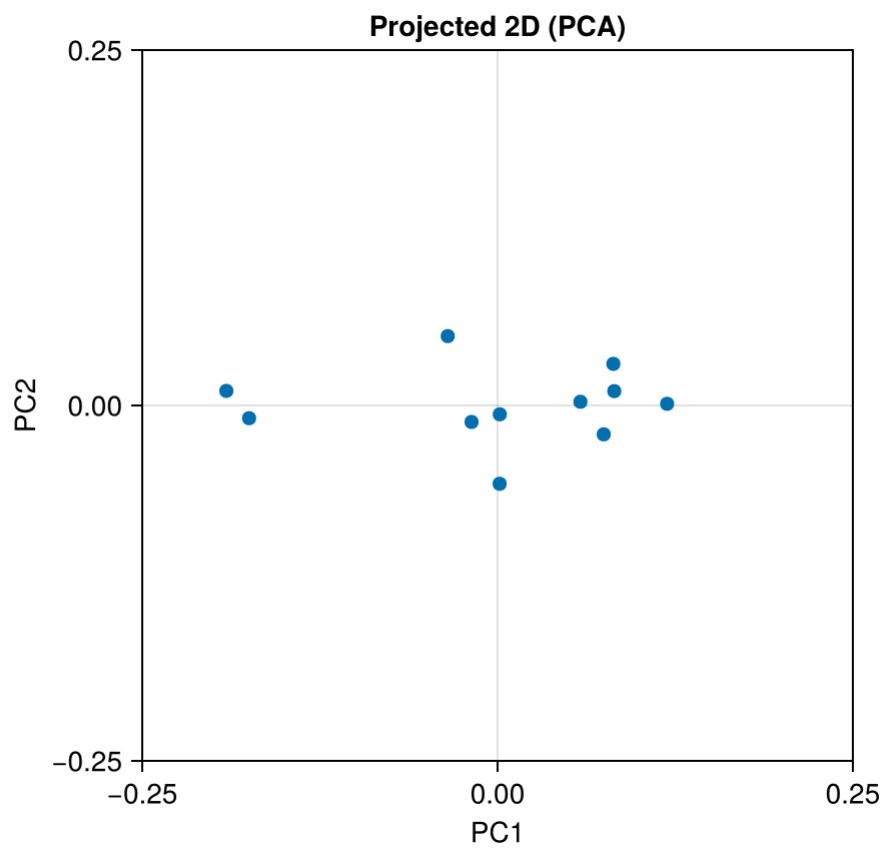
	PC1	PC2	PC3
1	0.0505687	0.00402319	-0.011892
2	0.0333734	0.00287332	0.0146519
3	0.0122172	0.00249094	0.000924746
4	0.0120326	0.00308355	0.00133029
5	0.00945411	0.0247719	-0.000315778
6	-0.017354	0.00131056	0.0113407
7	0.0504597	-0.00643142	-0.00454154
8	0.0244088	-0.000891694	0.00266848
9	0.0549223	-0.0042248	0.00817333

Importance of components:

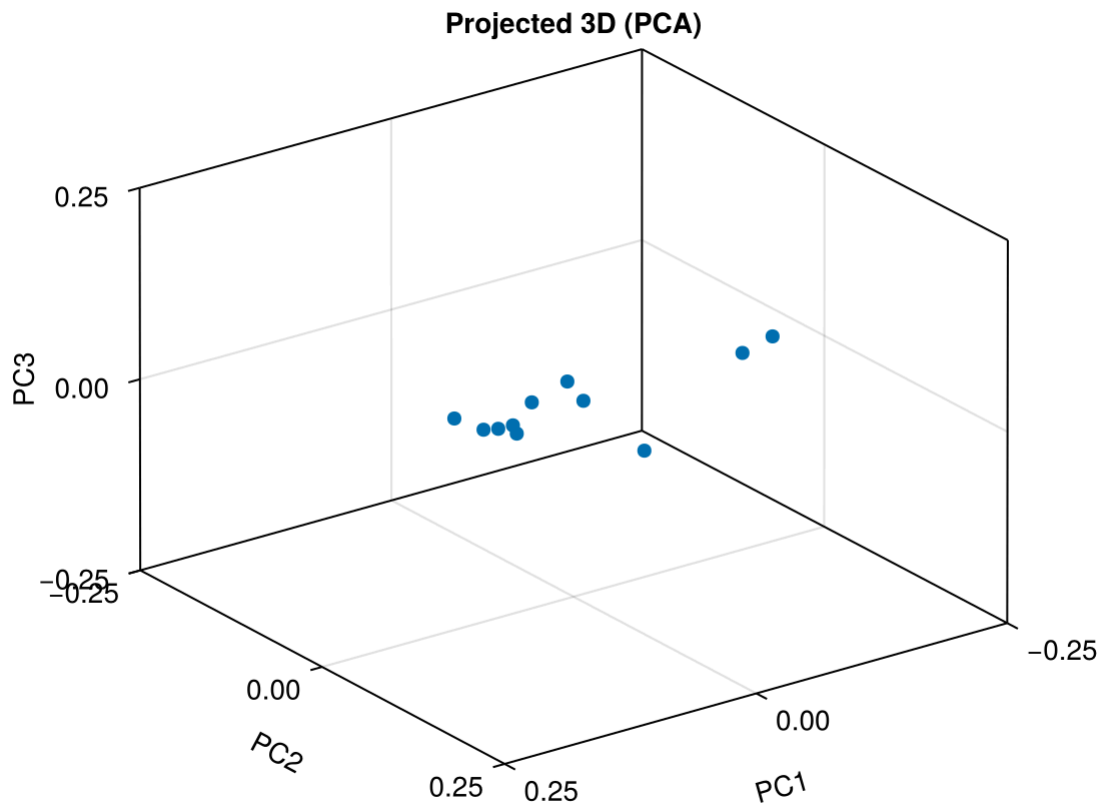
	PC1	PC2	PC3
SS Loadings (Eigenvalues)	0.010514	0.000715529	0.000581986
Variance explained	0.868227	0.0590872	0.0480594
Cumulative variance	0.868227	0.927314	0.975374
Proportion explained	0.890148	0.060579	0.0492728
Cumulative proportion	0.890148	0.950727	1.0

```
1 Mss = fit(PCA, Xs'; pratio=0.95)
```

```
1 X_projs = predict(Mss, Xs');
```

- And now using 3 PCs:



- Please consider that in [Covino et al. \(1995\)](#) the analysis there included more data from different sources that needed a proper homogenization we are not trying here.

Reference & Material

Material and papers related to the topics discussed in this lecture.

- [Covino et al. \(1995\)](#) - "Spectrophotometric indices and metal content of Galactic globular clusters"

Further Material

Papers for examining more closely some of the discussed topics.

- [Shlens \(2005\)](#) - "A Tutorial on Principal Component Analysis"

Credits

This notebook contains material obtained by <https://builtin.com/data-science/step-step-explanation-principal-component-analysis>.

Course Flow

	Previous lecture	Next lecture	Course Summary
notebook	Lecture about singular spectrum analysis	Lecture about singular spectrum analysis	Course Summary
html	Lecture about singular spectrum analysis	Lecture about singular spectrum analysis	Course Summary

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